CNSCC.160 Fundamentals of Communications Systems

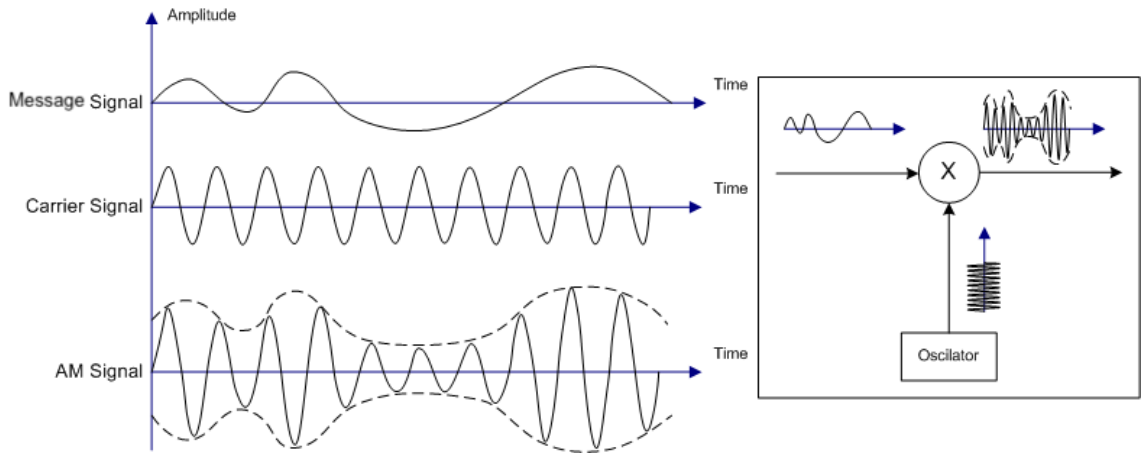
Lab 3: Analogue Modulation Techniques

1. Amplitude modulation

The purpose of this lab exercise is to gain familiarity with the concepts of amplitude modulation, you have studied in Lecture 3. It is expected that now you are familiar with the basic functions of MATLAB and the blocks of SIMULINK.

Conventional Amplitude Modulation

In amplitude modulation, the amplitude of a ***carrier*** signal is varied by the ***message***, whose frequency is lower than that of the carrier. In practice, the *carrier* may be high-frequency (HF) while the *message* is audio. Formally, AM is defined as *a system of modulation* in which the amplitude of the *carrier* is made proportional to the instantaneous amplitude of the modulating (message) voltage.



Let the *carrier* voltage and the message voltage, v_c and v_m , respectively, represented by

$$v_c = V_c \sin(2\pi f_c\, t) \qquad (1)$$

$$v_m = V_m\, sin\,(2\pi f_m t) \qquad (2)$$

Note that phase angle has been ignored in both expressions since it is unchanged by the amplitude modulation process. Its inclusion here would merely complicate the proceedings, without affecting the result. The amplitude of the modulated wave will be

$$A = V_c + v_m = V_c + V_m\, sin\,(2\pi f_m t)$$

$$= V_c(1 + m\, sin\,(2\pi f_m t)) \qquad (3)$$

The instantaneous voltage of the resulting amplitude modulated wave will be

$$v_{AM}(t) = A \sin(2\pi f_c\, t) = V_c\big(1 + m\, sin(2\pi f_m t))\big) \sin(2\pi f_c\, t) \qquad (4)$$

where $m = \frac{V_m}{V_c}$ (5)

m is also known as the modulation depth.

$$v_{AM}(t) = V_c \sin(2\pi f_c\, t) + m\, sin(2\pi f_m t)\ \sin(2\pi f_c\, t) \qquad (6)$$

Let us implement equation 6 in MATLAB and Simulink and observe the results.

Matlab Implementation

To use Matlab to simulate the AM procedure, edit a file, *am_lab.m*, , which should contains the following steps:

1) Generate the carrier sine wave – v_c
   a) Determine the parameters of the carrier sine wave by using the following commands

```
clear all
TD = 5; % the time duration in seconds
Frequency_carrier = 200; % the frequency of the carrier wave
Amplitude_carrier = 6; % the amplitude of the carrier wave
Frequency_sampling = 5000; % the sampling frequency
```

   b) Generate the carrier sine wave by using the following commands

```
% vc is the vector containing the values of the carrier wave and
t_vector is the time vector
t_vector = [1:TD*Frequency_sampling]/Frequency_sampling;
vc = Amplitude_carrier*sin(2* pi*Frequency_carrier*t_vector);
```

2) Generate the information bearing sine signal (message signal) – v_m

   a) Determine the parameters of the information bearing sine wave by using the following commands

```
% We use a sine wave to represent an information signal
Frequency_data = 20; % the frequency of the information signal
Amplitude_data = 3; % the amplitude of the information signal
```

   b) Generate the carrier sine wave by using the following commands

```
% vm is the vector containing the values of the information signal
and t_vector is the time vector
vm = Amplitude_data*sin(2* pi*Frequency_data*t_vector);
```

3) Generate the AM signal – z(t)

   a) Generate the AM signal by using the following command

```
vam =
(Amplitude_carrier+Amplitude_data*sin(2*pi*Frequency_data*t_vector)).
*sin(2*pi*Frequency_carrier*t_vector);
```

b) Generate the figures including the carrier signal, the modulating signal and the modulated signal by using the following commands

```
figure;
plot(t_vector,vc);
xlim([0,0.2]);
xlabel('Time (s)')
ylabel('Signal')
title('Carrier Wave')

figure;
plot(t_vector,vm);
xlim([0,0.2]);
xlabel('Time (s)')
ylabel('Signal')
title('Modulating Signal')

figure;
plot(t_vector,vam)
xlim([0,0.2]);
xlabel('Time (s)')
ylabel('Signal')
title('AM modulated signal')
```

4) Analyse the AM signal in the frequency domain

a) Generate the figure by using the following commands

```
function_spectrum_analysis(vam,Frequency_sampling)
xlim([100 300])
```

Note: xlim or ylim are used to set x-axis and y-axis limits to better show the figures. How to use them can be found here and here. You can also use **help xlim** or **help ylim** in the Command Window of MATLAB to find the helpful information.

Simulink Implementation

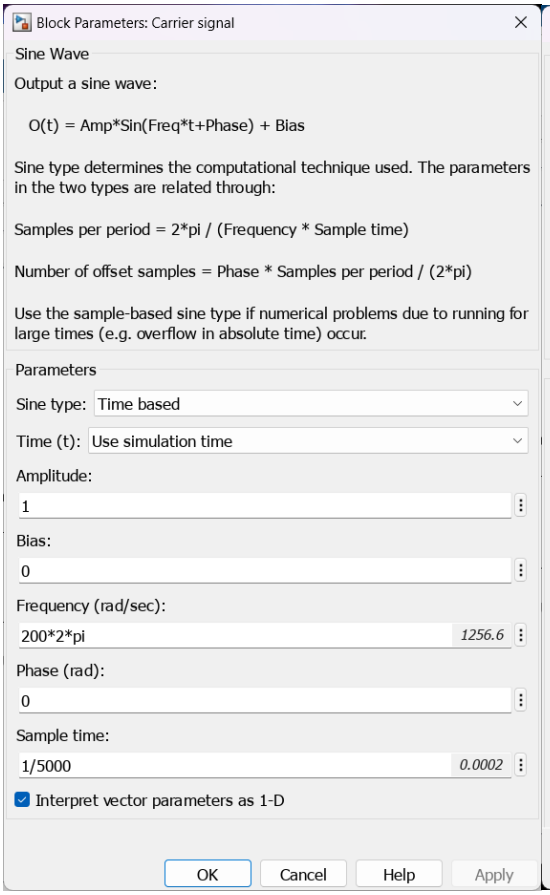
Let us now implement equation 6 in Simulink. Figure below illustrates the Simulink model for the system. You need Sine wave, product (multiplier), scope, adder and spectrum analyser blocks.

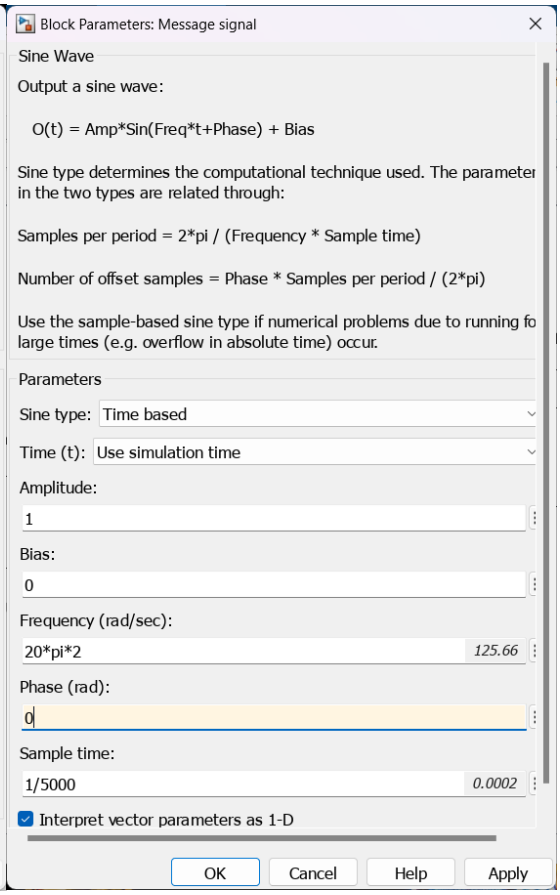
The parameters of simulation including message signal frequency, carrier frequency, and sampling rate are same as above

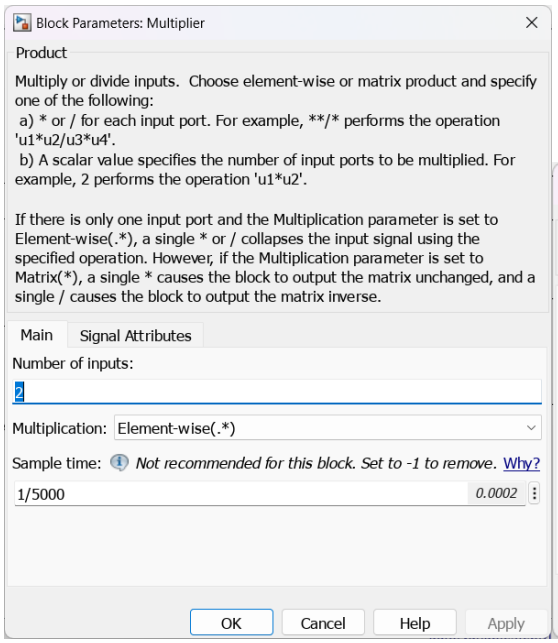
```
fc = 200;       %carrier freq
fm = 20;        %message freq
Fs = 5000;      %sample rate
Ts=1/Fs;        %sample time
Vc= 1;          %carrier amplitude
Vm=1;           %message amplitude
m= 1            %modulation depth
```

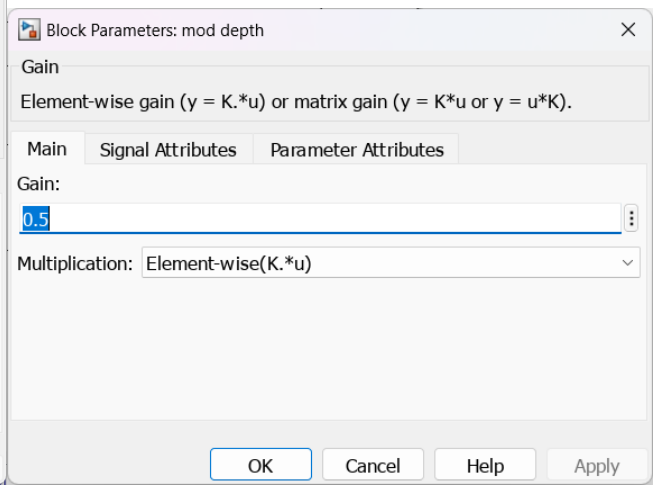
Carrier Signal= Simulink->source->Sine Wave block

Message Signal= Simulink->source->Sine Wave block



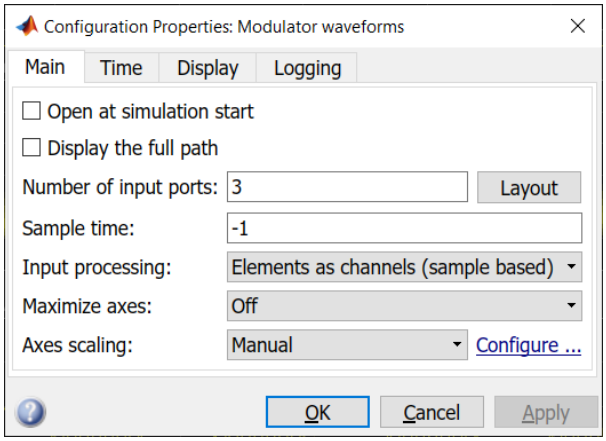






Multiplier – **Product Block = Simulink->math->product**

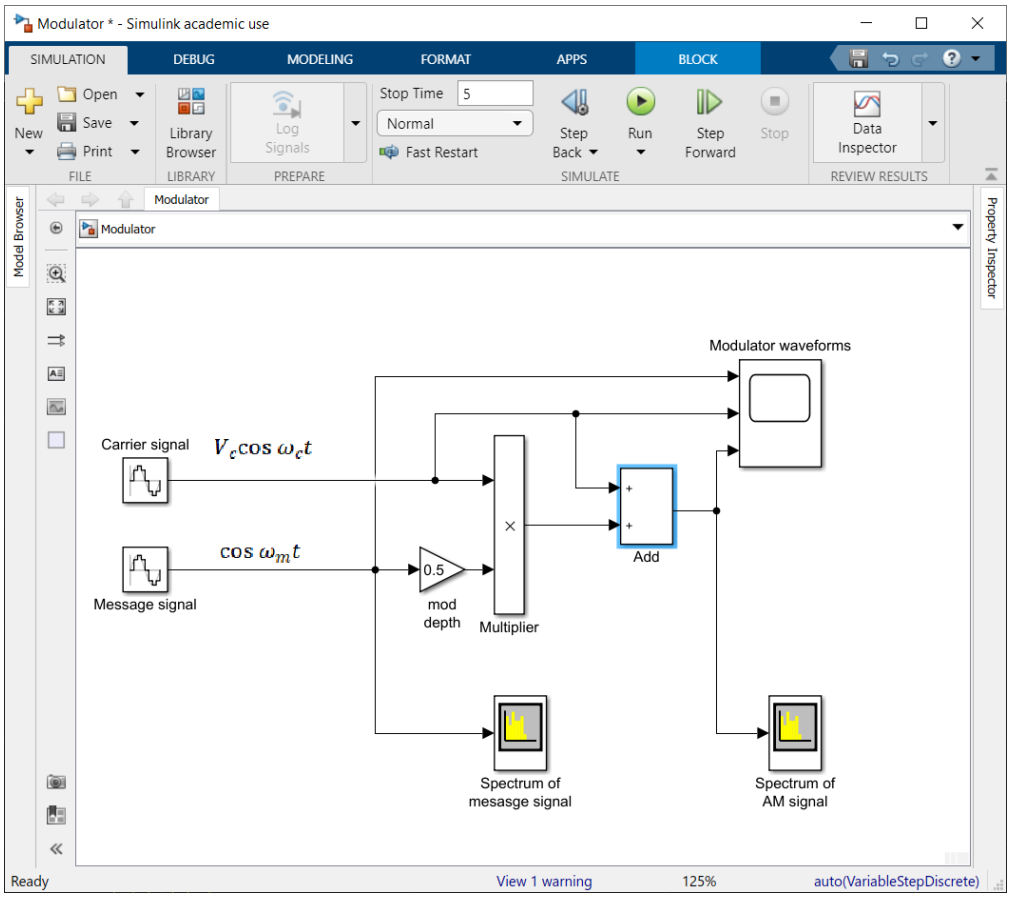
Mod Depth- **Gain Block = Simulink->math-> Gain**



Scope = Simulink->sink->scope

Spectrum analyser = DSP System ToolBox > sink-> Spectrum Analyser

Spectrum analyser



TASK 1: Use the provided m-file, i.e., function_spectrum_analysis.m and am_lab.m. Make the following changes and observe what happens in the figures.

```
%Analyse the AM signal in the frequency domain
function_spectrum_analysis(vam,Frequency_sampling)
xlim([100 300])
```

- **Increase or decrease the frequency of the carrier signal (the variable termed as "Frequency_carrier"). Comment on the changes of the carrier signals in the time and spectrum domains.**

- **Increase or decrease the frequency of the information bearing signal (the variable termed as "Frequency_data"). Comment on the changes of the information signals in the time and spectrum domains.**

- **Change the amplitudes of the carrier signal and the information signal. Comment on the changes of the AM signals in the time and spectrum domains.**
- **Change the frequencies of the carrier signal and the information signal. Comment on the changes of the AM signals in the spectrum domains.**
- **Do experiment with your Simulink Model**

TASK 2: In practice, the generated AM signals can be corrupted by noise. A typical example is additive white Gaussian noise (AWGN), and Matlab provides a simple way to add noise to signals. Particularly, given *vam* as the AM signal, you can use awgn(vam, SNR, 'measured') to add AWGN, where SNR denotes the signal to noise ratio and reflects how much noise will be added. e.g.

```
vam_noisy= awgn(vam, 1, 'measured');
figure;
plot(t_vector,vam_noisy)
xlim([0,0.2]);
xlabel('Time (s)')
ylabel('Signal')
title('AM modulated signal with additive noise')
%Analyse the AM signal in the frequency domain
function_spectrum_analysis(vam_noisy, Frequency_sampling)
xlim([100 300])
```

Here, when 'measured' is used, the signal level of vam will be computed to determine the appropriate noise level based on the value of SNR. Since in the awgn function, the parameter SNR is in dB, it can be a positive or negative value. Based on the AM signal generated previously, please carry out the following experiments.

- **Increase or decrease the value of SNR, and comment on the changes of the corrupted AM signals in the time domain.**

- **Increase or decrease the value of SNR, and comment on the changes of the corrupted AM signals in the spectrum domain.**